

Direct manipulation systems offer the satisfying experience of operating on visible objects. The computer becomes transparent, and users can concentrate on their tasks.

Direct Manipulation: A Step Beyond Programming Languages

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Leibniz sought to make the form of a symbol reflect its content. "In signs," he wrote, "one sees an advantage for discovery that is greatest when they express the exact nature of a thing briefly and, as it were, picture it; then, indeed, the labor of thought is wonderfully diminished."

Frederick Kreiling, "Leibniz,"
Scientific American, May 1968

Certain interactive systems generate glowing enthusiasm among users—in marked contrast with the more common reaction of grudging acceptance or outright hostility. The enthusiastic users' reports are filled with positive feelings regarding

- mastery of the system,
- competence in the performance of their task,
- ease in learning the system originally and in assimilating advanced features,
- confidence in their capacity to retain mastery over time,
- enjoyment in using the system,
- eagerness to show it off to novices, and
- desire to explore more powerful aspects of the system.

These feelings are not, of course, universal, but the amalgam does convey an image of the truly pleased user. As I talked with these enthusiasts and examined the systems they used, I began to develop a model of the features that produced such delight. The central ideas seemed to be visibility of the object of interest; rapid, reversible, incremental actions; and replacement of complex command language syntax by direct manipulation of the object of interest—hence the term "direct manipulation."

Examples of direct manipulation systems

No single system has all the attributes or design features that I admire—that may be impossible—but those described below have enough to win the enthusiastic support of many users.

Display editors. "Once you've used a display editor, you'll never want to go back to a line editor. You'll be spoiled." This reaction is typical of those who use full-page display editors, who are great advocates of their systems over line-oriented text editors. I heard similar comments from users of stand-alone word processors such as the Wang system and from users of display editors such as EMACS on the MIT/Honeywell Multics system or "vi" (for visual editor) on the Unix system. A beaming advocate called EMACS "the one true editor."

Roberts¹ found that the overall performance time of display editors is only half that of line-oriented editors, and since display editors also reduce training time, the evidence supports the enthusiasm of display editor devotees. Furthermore, office automation evaluations consistently favor full-page display editors for secretarial and executive use.

The advantages of display editors include

Display of a full 24 to 66 lines of text. This full display enables viewing each sentence in context and simplifies reading and scanning the document. By contrast, the

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